

Deliverable 4

Collin Brueggeman

```
In [1]: import pandas as pd  
import numpy as np
```

```
In [2]: heart_data = pd.read_csv('healthcare-dataset-stroke-data.csv')  
heart_data_cleaned = pd.read_csv('healthcare-dataset-stroke-data_cleaned.csv')
```

Problem Statement and Project Context

Problem Statement and Project Purpose

Stroke is one of the leading causes of death and disability worldwide. Early prediction of stroke risk can lead to preventive healthcare interventions and reduce healthcare costs. Existing research shows that machine learning models can effectively identify high-risk individuals using health and lifestyle indicators.

The objective of this project is to use predictive analytics to develop a model that predicts the likelihood of stroke occurrence based on patient health attributes. This application of predictive analytics has practical significance for healthcare providers seeking to provide preventive care resources more effectively and can also help individuals make better health decisions.

My problem statement is supported by this systematic review F. Asadi, M. Rahimi, A. H. Daechini, and A. Paghe, "The most efficient machine learning algorithms in stroke prediction: A systematic review," Health Science Reports, vol. 7, no. 10, Oct. 2024, doi: <https://doi.org/10.1002/hsr2.70062>. as it highlights stroke being a major issue and that machine learning/predictive analytics is a growing method of early risk predictions.

The objective of my project is supported by E. Alanazi, A. Abdou, and J. Luo, "Predicting risk of stroke from lab tests using machine learning algorithms (Preprint)," JMIR Formative Research, Aug. 2020, doi: <https://doi.org/10.2196/23440>. which details models built on health attributes to predict stroke risk which validates my approach.

This modeling goal is relevant towards both advancing the research field relating to using predictive analytics in the healthcare field and helping industry by providing insight and advancements in tools to save consumers and businesses/service providers money while maintaining confidence in their abilities. My modeling choices are supported by the systematic review mentioned above where 58 studies were reviewed for what models were used and the results from them.

Critical Prior Work

Dataset Characteristics and Acquisition

The research utilized the "Stroke Prediction Dataset" sourced from Kaggle which comprises 5110 patient records. The dataset contains 12 attributes including the binary target variable stroke. Feature vectors consisted of demographic details such as gender and age alongside clinical indicators including hypertension, heart disease, average glucose level, and body mass index (BMI). Initial inspection revealed a mix of categorical and numerical data types requiring substantial preprocessing.

Data Cleaning

Some of the data cleaning strategies deployed were as follows:

- **Missing Value Imputation:** The dataset contained 201 missing entries for BMI which were remedied using median imputation
- **Outlier Management:** Outliers within the BMI feature were mitigated by capping values at the 99th percentile to preserve distribution integrity while reducing noise.
- **Feature Encoding:** Categorical variables underwent transformation where binary encoding was applied to gender, marital status, and residence type. One-hot encoding was utilized for multi-class nominal variables such as work type and smoking status.

Exploratory Data Analysis Insights

Key insights gained from the EDA portion of this project include:

- **Class Imbalance:** A severe class imbalance was observed where stroke occurrences constituted merely 4.87% of the total observations.
- **Correlation Analysis:** Age exhibited the strongest positive linear association with stroke risk (approx 0.25) followed by hypertension (approx 0.13) and heart disease (approx 0.13).
- **Principal Component Analysis (PCA):** PCA demonstrated that the variance in the dataset was primarily driven by average glucose levels (PC1), age (PC2), and BMI (PC3).

Data Preparation Summary

Missing values:

- Missing values such as the ones seen in the "bmi" feature were imputed using median imputation [5].
- For "Unknown" in "smoking_status", it was encoded as a sperate category [6].

Outliers:

- Outliers (bmi) were capped to the 99th percentile.

Irrelevant Fields:

- Irrelevant fields were dropped (such as "id").

Encoding:

- Binary encoding for two-level variables: "gender", "ever_married", and "residence_type"
- One-hot encoding for multi-level variables: "work_type" and "smoking_status"

Feature Scaling:

- Feature Scaling was done using StandardScaler() to prevent features with larger ranges of values to dominate distance-based models used such as SVM.

Class imbalance:

- SMOTE was applied to the feature set to remedy the large class imbalance found within the dataset (only 4.87% stroke).

All of the above techniques can be found in our book [5] or the book by Kuhn & Johnson [6].

Before

In [3]: heart_data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   id                    5110 non-null   int64
1   gender                5110 non-null   object
2   age                  5110 non-null   float64
3   hypertension          5110 non-null   int64
4   heart_disease         5110 non-null   int64
5   ever_married          5110 non-null   object
6   work_type             5110 non-null   object
7   Residence_type        5110 non-null   object
8   avg_glucose_level     5110 non-null   float64
9   bmi                   4909 non-null   float64
10  smoking_status        5110 non-null   object
11  stroke                5110 non-null   int64
dtypes: float64(3), int64(4), object(5)
memory usage: 479.2+ KB
```

After

In [4]: heart_data_cleaned.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 17 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Unnamed: 0                            5110 non-null   int64
1   gender                                5110 non-null   int64
2   age                                    5110 non-null   float64
3   hypertension                          5110 non-null   int64
4   heart_disease                        5110 non-null   int64
5   ever_married                         5110 non-null   int64
6   Residence_type                       5110 non-null   int64
7   avg_glucose_level                    5110 non-null   float64
8   bmi                                   5110 non-null   float64
9   stroke                               5110 non-null   int64
10  work_type_Never_worked                5110 non-null   bool
11  work_type_Private                     5110 non-null   bool
12  work_type_Self-employed               5110 non-null   bool
13  work_type_children                    5110 non-null   bool
14  smoking_status_formerly smoked        5110 non-null   bool
15  smoking_status_never smoked           5110 non-null   bool
16  smoking_status_smokes                 5110 non-null   bool
dtypes: bool(7), float64(3), int64(7)
memory usage: 434.3 KB
```

Justification

The above cleaning and encoding methods were taken from the below linked study that looked at 58 different research papers that tackled the task of predicting strokes from medical data. I chose to use these best practices to ground my work in what has been the standard in this area of predictive modeling. BMI outliers needed to be dealt with as the most extreme measurements of BMI I could find were in the 50-60 range. After researching into the glucose levels I decided to keep the original values to take into the fact that the outliers could have to do with people having diabetic conditions. I decided to binary encode the categorical features that only had 2 unique values and dummy encode the rest as that was common practice I found in my reading.

A. A. Soladoye, N. Aderinto, M. R. Popoola, I. A. Adeyanju, A. Osonuga, and D. B. Olawade, "Machine learning techniques for stroke prediction: A systematic review of algorithms, datasets, and regional gaps," *International Journal of Medical Informatics*, vol. 203, p. 106041, Jul. 2025, doi: <https://doi.org/10.1016/j.ijmedinf.2025.106041>.

Model Development and Results

Logistic Regression:

- Seen as a strong baseline model for medical classification problems.
- Provides interpretable coefficients that are valuable in the healthcare field for identifying patient attributes that influence conditions.

- Assumptions:
 - Features have a linear relationship with the log-odds of the outcome.
 - Predictor features are not highly multicollinear.
 - Data points are independent.
- Justification:
 - Logistic Regression is suitable for my problem because it is interpretable and performs well when the relationship is roughly linear which commonly happens in medical datasets. Lastly, many of the studies I looked at from the systematic review found in the problem statement use Logistic Regression as a baseline model in their studies.

Logistic Regression Metrics

- Accuracy: 0.855
- Precision: 0.835
- Recall: 0.885
- F1 Score: 0.859

Support Vector Machine (SVM)

- Kernel functions allow SVM to capture complex decision boundaries around stroke and non-stroke cases.
- Assumptions:
 - The dataset is separable in some feature space.
 - Patterns can be captured by maximizing the margin between classes.
- Justification:
 - SVM creates robust boundaries and has been seen to be less sensitive to class imbalance when using correct hyperparameters. Many of the studies I looked at had SVM as one of the best performing models which supports its use in this problem.

SVM Metrics

- Accuracy: 0.871
- Precision: 0.855
- Recall: 0.893
- F1 Score: 0.873

Decision Tree

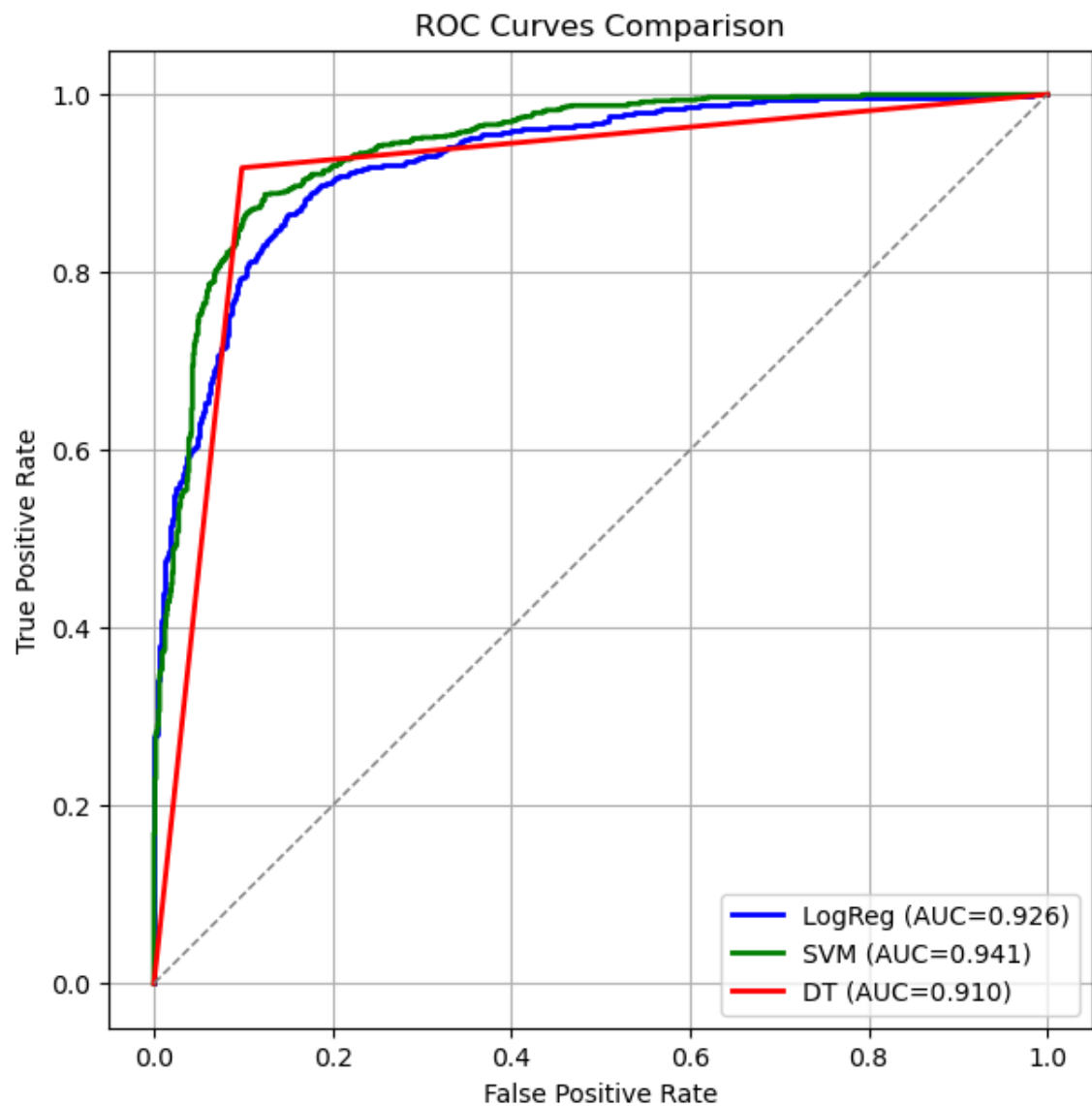
- Decision Trees are known for being interpretable and can reveal clear rule-based pathways that even non-technical people such as doctors can understand. Assumptions:
 - Decision boundaries will be able to be expressed as hierarchical if-then rules.
 - Enough data is present to prevent overfitting.
- Justification:

- Decision Trees are perfect for this use case because they are able to model complex relationships and have a high degree of interpretability which is important in a life-or-death situation that the healthcare field often is. Literature also supports the use of Decision Trees as they were among the most commonly tested in the studies I read.

Decision Tree Metrics

- Accuracy: 0.910
- Precision: 0.903
- Recall: 0.918
- F1 Score: 0.910

ROC/AUC



Best Hyperparameters post GridSearchCV

Logistic Regression

- F1 Score: 0.8497217707899949
- Best Parameters: {'C': 0.1, 'max_iter': 500, 'penalty': 'l1', 'solver': 'liblinear'}

SVM

- F1 Score: 0.8846352717110854
- Best Parameters: {'C': 20, 'degree': 2, 'gamma': 'scale', 'kernel': 'rbf'}

Decision Tree

- F1 Score: 0.9096043785980699
- Best Parameters: {'criterion': 'log_loss', 'max_depth': 30, 'min_impurity_decrease': 0, 'min_samples_leaf': 1, 'min_samples_split': 2}

Side by Side Comparison (Tuned Models)

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.857	0.833	0.891	0.861
SVM	0.887	0.859	0.927	0.891
Decision Tree	0.914	0.908	0.922	0.915

Best Model: Decision Tree

The Decision Tree model has the best results overall, as it produces the best accuracy (.914) and F1 scores (.915) which indicate good consistency and fairness of classification results of both positive and negative stroke-risk classifications. The Decision Tree model is able to better detect non-linear interaction effects among health characteristics than the linear Logistic Regression model and slightly outperforms the SVM model in terms of precision, with a similar level of recall to the SVM model. A high degree of precision, or few false alarms, and a high degree of sensitivity, or few false negatives, will make the model useful for early screening purposes where there are serious consequences associated with either missing detection or producing too many false alarms.

Decision Trees offer additional value beyond just their predictive strength. The SVM model offers strong predictive power, but offers less interpretability of the predictive process than the Decision Tree model. Decision Trees offer a clear, understandable structure to healthcare professionals split by split and therefore support ethically responsible model use, as patients and providers, should be able to question the predictive process of any model that is used to guide decisions that impact patient care. Logistic Regression offers some interpretation capabilities, but limits itself through its assumption of linearity in the relationship between stroke risk factors.

There are trade-offs to consider here. If not properly tuned, Decision Trees can fit the training data too closely and become overly sensitive to small variations in the data. Also, even when well-tuned, the boundary between high and low stroke-risk will vary based upon the specific values of the input variables. SVM models are very powerful, but lack transparency and therefore cannot be inspected or questioned in the same way as Decision Trees. Lastly, Logistic Regression models may perform poorly in cases where the relationship between the predictor variables and the outcome variable includes important feature interactions. With all the above trade-offs and benefits in mind, the Decision Tree model appears to offer the best overall combination of performance, interpretability and practical responsibility in this case.

Lessons Learned

Highlights

The primary success of this project was the development of a highly effective Decision Tree model, which achieved an accuracy of 0.914 and an F1 score of 0.915. This model successfully balanced precision (0.908) and recall (0.922), a critical requirement for medical screening tools where missing a positive stroke case (false negatives) can lead to severe consequences. The project successfully implemented rigorous data cleaning protocols based on systematic reviews, specifically regarding the handling of BMI outliers and missing values. By capping BMI outliers at the 99th percentile and imputing missing values with the median, the project preserved data integrity without allowing extreme physiological impossibilities to skew the model.

Lowlights

The most significant challenge encountered was the management of class imbalance. While the application of the Synthetic Minority Over-sampling Technique (SMOTE) allowed the models to learn from the minority class, a massive mismatch between scholarly work and best practices was found. The oversampling was applied to the entire dataset before the train-test split. This likely introduced "data leakage," where synthetic variations of validation data points leaked into the training set, potentially inflating the evaluation metrics and making the results appear more robust than they would be in a real-world clinical setting. I touched on this at the end of Deliverable 3 and about how some of the studies I looked at could potentially lose some credibility based on their approach to SMOTE versus the best practice outlined in our Data Mining for Business Analytics book.

Changes I Would Implement on a New Project

I would choose a different dataset that contained more data points and features to give more room to experiment with feature engineering and hopefully provide even more predictive power. The next thing I would change is how I implemented SMOTE. In future

projects I would like to attempt to use best practices outlined in the book such as applying SMOTE to just the training set instead of the entire dataset that I witnessed in the studies I read for this project.

Deployment Plan

The decision tree outlined in this project as the best model could be deployed using the DEPLOYR framework.

The model is deployed to the inference engine, and exposed as a REST API using DEPLOYR-serve package. Inference is then triggered which spawns an HTTPS request from the electronic healthcare records that are being fed to the exposed model. The request results in the collection of a feature vector from the electronic medical record's database using REST APIs maintained by the electronic medical records vendor. Inference is performed on the real-time data and routed back to the electronic medical records to close the loop with end-users and integrate into workflow. Inferences and relevant metadata are saved to the inference store. These are then consumed by the DEPLOYR-dash monitoring software that continuously tracks model performance via a dashboard.

The deployed model is able to be monitored over time to ensure performance doesn't drift. If the model begins to drift it can be taken down, retrained in house with the electronic medical records at hand, and then sent back into inference duty. The model would need to be closely monitored to examine to see if there are any biases in specific groups of people that arise. The privacy of medical data is secured by closing the loop mentioned above and not allowing data to leave in-house systems.

Source:

Conor K Corbin, Rob Maclay, Aakash Acharya, Sreedevi Mony, Soumya Punathanam, Rahul Thapa, Nimesh Kotecha, Nigam H Shah, Jonathan H Chen, DEPLOYR: a technical framework for deploying custom real-time machine learning models into the electronic medical record, Journal of the American Medical Informatics Association, Volume 30, Issue 9, September 2023, Pages 1532–1542, <https://doi.org/10.1093/jamia/ocad114>

References

[1] F. Asadi, M. Rahimi, A. H. Daechini, and A. Paghe, "The most efficient machine learning algorithms in stroke prediction: A systematic review," Health Science Reports, vol. 7, no. 10, Oct. 2024, doi: <https://doi.org/10.1002/hsr2.70062>.

[2] A. Hassan, S. Gulzar Ahmad, E. Ullah Munir, I. Ali Khan, and N. Ramzan, "Predictive modelling and identification of key risk factors for stroke using machine learning," Scientific Reports, vol. 14, no. 1, p. 11498, May 2024, doi: <https://doi.org/10.1038/s41598-024-61665-4>.

- [3]S. Dev, H. Wang, C. S. Nwosu, N. Jain, B. Veeravalli, and D. John, "A predictive analytics approach for stroke prediction using machine learning and neural networks," arXiv.org, Mar. 01, 2022. <https://arxiv.org/abs/2203.00497>
- [4]L. Ismail and Huned Materwala, "From Conception to Deployment: Intelligent Stroke Prediction Framework using Machine Learning and Performance Evaluation," arXiv, pp. 1–7, Aug. 2022, doi: <https://doi.org/10.1109/coins54846.2022.9854961>.
- [5]B. Peter, G. Shmueli, P. Gedeck, and N. R. Patel, DATA MINING FOR BUSINESS ANALYTICS : concepts, techniques and applications in python. S.L.: Wiley-Blackwell, 2020.
- [6]M. Kuhn and K. Johnson, Feature Engineering and Selection. CRC Press, 2019.